THE STRICT TRANSFORM OF A MODULE FREE OFF THE FITTING IDEAL IS LOCALLY FREE OF THE SAME RANK

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The Fitting ideal of the strict transform under the blowup. Lemma \ref{lemma-blowup-fitting-ideal}, cited at proof lines 6211-6212. Gives the top Fitting ideal as the unit ideal.
- A finite module is finite locally free of rank k exactly when the k-th Fitting ideal is the unit ideal and the (k-1)-st vanishes. Lemma \ref{lemma-fitting-ideal-finite-locally-free}, cited at proof lines 6212-6213 and 6225-6226. Turns the two Fitting ideal computations into local freeness.
- The blowup algebra sits inside its localisation at an element of the ideal. Algebra, Lemma \ref{algebra-lemma-affine-blowup}, cited at proof lines 6214-6216. Lets the vanishing of the Fitting ideal be checked after inverting.
- Formation of Fitting ideals commutes with base change. Lemma \ref{lemma-fitting-ideal-basics}, cited at proof lines 6219-6221. Identifies the Fitting ideal of the localised transform.

Source reference. Original lemma 0CZN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16283–16283. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0CZM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16282–16282. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

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Source reference. Original lemma 07ZD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10143–10143. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07Z3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10133–10133. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07ZA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10140–10140. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00NX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 875–875. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let M be a finite R-module. Let $k \geq 0$ and $I = Fit_k(M)$. Assume that $M_\mathfrak{p}$ is free of rank k for every $\mathfrak{p} \notin V(I)$. Then for every $a \in I$ with $R' = R[\frac{I}{a}]$ the strict transform*

$$M' = (M \otimes_R R')/a\text{-power torsion}$$

is locally free of rank k.

Proof. By Lemma 0CZM we have $\mathrm{Fit}_k(M') = R'$. By Lemma 07ZD it suffices to show that $\mathrm{Fit}_{k-1}(M') = 0$. Recall that $R' \subset R'_a = R_a$, see Algebra, Lemma 07Z3. Hence it suffices to prove that $\mathrm{Fit}_{k-1}(M')$ maps to zero in $R'_a = R_a$. Since clearly $(M')_a = M_a$ this reduces us to showing that $\mathrm{Fit}_{k-1}(M_a) = 0$ because formation of Fitting ideals commutes with base change according to Lemma 07ZA part (3). This is true by our assumption that M_a is finite locally free of rank k (see Algebra, Lemma 00NX) and the already cited Lemma 07ZD. □